Mohamad Mustafa Abdulkadir

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EDUCATION

United Arab Emirates University Jan. 2021 – May 2025
Bachelor of Science in Computer Science, ***GPA: 3.97 / 4.00*** Al Ain, UAE

SKILLS

Languages: Python, TypeScript, JavaScript, SQL
Frameworks & Libraries: React, Next.js
AI & Machine Learning: LLMs (Training, Fine-tuning, RAG), OpenAI/Anthropic APIs, Docling
Cloud, Databases & Tools: AWS, PostgreSQL, Snowflake, Qdrant, Docker, Nginx, PM2, Postman

EXPERIENCE

Mansoura Oct. 2025 – Present
Software Engineer - Data & AI Abu Dhabi, UAE

- Built and launched a full-stack platform using React, TypeScript, AWS, PostgreSQL, and Snowflake.
- Developed authentication, role-based access control, backend APIs, document storage and generation, and scheduled cloud workflows.
- Automated operational reporting and integrated KPI forecasting.
- Developed an agentic AI workflow that matches unidentified rental payments from bank statements to tenants using their payment history.

ADIA Lab June 2025 – Sept. 2025
Machine Learning Research Intern Abu Dhabi, UAE

- Trained a transformer-based encoder to predict in-hospital mortality from time-series converted EHR data, managing compute-intensive experiments and systematically tracking model performance.
- Built and optimized an end-to-end pipeline for transforming longitudinal EHR records into model-ready time series and clinical narratives, reducing projected processing time from 193.4 hours to 18 hours.
- Derived patient-level explanations of the encoders mortality predictions using attention- and gradient-based attribution, then used them alongside clinical narratives to fine-tune a transformer decoder for grounded report generation.

e& Jan. 2025 – Apr. 2025
AI Intern Dubai, UAE

- Worked on integrating an AI-powered content recommendation engine into the Starz On Business platform.
- Built a solution using product guides and FAQs to automate tier-one customer support cases.

ATRC - STEM Youth Mentorship Program Nov. 2023 – Sept. 2024
Research Assistant Abu Dhabi, UAE

- Researched non-stationary contextual-bandit algorithms and large language models for recommender systems.
- Introduced change-point detectors to contextual bandit algorithms to improve adaptation in stochastic environments.

United Arab Emirates University May 2023 – Jan. 2025
Research Assistant Al Ain, UAE

- Developed an AI-powered educational chatbot for students with special needs and implemented machine learning-based techniques to enhance task classification in mobile crowdsensing platforms.
- Engineered a task-allocation framework using a combination of machine learning models and filtering algorithms, utilizing synthetic training data generation to improve classification accuracy.

PROJECTS

AwesomeRAG

- Full-stack document RAG workspace supporting document uploads, OCR, vector retrieval, intent routing, and grounded responses with source attribution.
- Used locally deployed Docling OCR and EmbeddingGemma-300M, with Gemini as an optional cloud fallback and Qdrant for vector storage.

German Power Day-Ahead Forecasting Pipeline

- Built an end-to-end pipeline integrating 10 ENTSO-E and EEX datasets, with automated ingestion, normalization, deterministic and LLM-generated quality checks, and hourly feature engineering.
- Benchmarked forecasting models using walk-forward validation, achieving a best MAE of 13.65 EUR/MWh. Translated hourly forecasts into weekly and monthly baseload, peak, and off-peak fair-value bands for market-signal analysis.

OmniSpaces

- Platform for organizing and semantically searching multimodal assets (text, image, video, audio).
- Developed using Next.js, Supabase, AWS S3, and Qdrant, using Gemini 3.1 Flash-Lite for conversational RAG.

NutriVision

- Mixed Reality diet-tracking application that identifies food, estimates item weight, and calculates nutritional content.
- Trained deep learning models for weight estimation and fine-tuned YOLOv11 for real-time detection; placed third in the UAEU Innovation Challenge and presented the project at Expand North Star Dubai 2024.

PUBLICATIONS

- **DoS-based Fake Task Injection for Disrupted Sensing**, AICCSA, 2025.
- **Vault-PMS: A Vault-Based Password Management System**, IWCMC, 2024.

OPEN-SOURCE CONTRIBUTIONS

Deep-ML

- Contributed LeetCode-style problems focused on machine learning and linear algebra.